

What's wrong with WebSocket APIs? Unveiling vulnerabilities in WebSocket APIs.

What's wrong with WebSocket APIs? Unveiling vulnerabilities in WebSocket APIs. -

@speakerdeck, Mikhail Egorov, Speaker Deck.

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Slides from Hactivity 2019 conference - <https://hactivity.com/index.php/presentations/>.

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Mikhail Egorov

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Transcript

What's wrong with w/ WebSocket API? Unveiling vulnerabilities in WebSocket

APIs Mikhail Egorov / @0ang3el #HACKTIVITY2019

whoami 2 Security researcher / full-time bug hunter

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<https://chybeta.github.io/2018/04/07/spring-messaging-Remote-Code-Execution-%E5%88%86%E6%9E%90-%E3%80%90CVE-2018-1270%E3%80%91/>
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4 WebSocket protocol essentials

WebSocket protocol – RFC 6455 5 Efficient two-way communication

protocol WebSocket is stateful (HTTP is stateless) Two main parts: handshake and data transfer

WebSocket protocol – RFC 6455 6 Extensibility: subprotocols and

extensions Subprotocols
<https://www.iana.org/assignments/websocket/websocket.xml#subprotocol-name> Wamp
Stomp Soap ...

WebSocket protocol – RFC 6455 7 Extensibility: subprotocols and

extensions Extensions <https://www.iana.org/assignments/websocket/websocket.xml#extension-name> permessage-deflate bbf-usb-protocol

-

WebSocket protocol – RFC 6455 8 Origin-based security model

(Browser clients) No authentication Client must do client-to-server masking

-

WebSocket protocol support 9 Major web browsers Web

servers / Proxies Apache httpd, Nginx, IIS, ... HAProxy, Traefik, Varnish, Envoy, ... Cloud providers WebSocket API (api gateways) WebSocket proxying (load balancers)

-

WebSocket handshake 10 Upgrade request Base64(Random nonce) Protocol version Required

HTTP version

-

WebSocket handshake 11 Required status code BASE64(SHA1(Sec-WebSocket-Key || CONST))

-

WebSocket data transfer 12 \x00 – continuation frame \x01 –

text frame \x02 – binary frame \x08 – close frame \x09 – ping \x0A – pong other values are reserved

-

WebSocket data transfer - masking 13 Masking key is

32-bit long passed inside frame Client must send masked data $MASKED = MASK \oplus DATA$ ($\oplus$ - XOR) Mechanism protects against cache poisoning and smuggling attacks

-

14 Cross-Site WebSocket Hijacking

-

WebSocket security for Web Browser 15 SOP doesn't work

for WebSocket in web browser Read from WebSocket cross-origin Write to WebSocket cross-origin Header Origin should be checked on handshake step (origin-based security model)

-

CSWSH 16 Cookies are used to authenticate upgrade request

Header Origin isn't checked or checked poorly

-

CSWSH 17 CORS tricks from @albinowax are applicable to

WebSocket <https://portswigger.net/research/exploiting-cors-misconfigurations-for-bitcoins-and-bounties> Null origin Pre-domain wildcard Post-domain wildcard ...

-

CSWSH - Null origin 18 `nullorigin.html <iframe src="data:text/html, <script>const`

`socket = new WebSocket('wss://example.com'); </script>"></iframe>`

-

CSWSH 19 Playground <https://portswigger.net/web-security/websockets/cross-site-websocket-hijacking>

-

CSWSH - template for attack 5

-

Demo 5

-

22 Authentication / IDOR issues

-

Authentication 23 WebSocket protocol doesn't offer authentication Developers

have to roll out their own AuthN It's secure to check AuthN only during handshake Common secure implementations Session cookies Tokens

-

Broken authentication - Case 1 24 Some ID /

GUID is required in Upgrade request Guess ID Leak GUID (minor IDOR, ...)

-

Broken authentication - Case 2 25 No authentication during

handshake step Some ID / GUID required in API messages Guess ID Leak GUID (minor IDOR, ...)

-

Broken authentication – Case 2 26 Exposing GraphQL subscriptions

w/o AuthN <https://github.com/righettod/poc-graphql#subscriptions- websocket-endpoint-default-enabling> Path /subscriptions

-

Insecure Direct Object Reference issues 27 Strong authentication during

handshake step Some ID / GUID required in API messages Guess ID Leak GUID (minor IDOR, ...)

-

28 Smuggling through WebSocket

-

Reverse proxying WebSocket connection 29 Client Frontend Reverse proxy Backend

/socket.io/ Public WebSocket API

-

Reverse proxying WebSocket connection 30 Client Frontend Reverse proxy Upgrade

request Upgrade request Backend /socket.io/

-

Reverse proxying WebSocket connection 31 Client Frontend Reverse proxy Upgrade

request Upgrade request HTTP/1.1 101 HTTP/1.1 101 Backend /socket.io/

-

Reverse proxying WebSocket connection 32 Client Frontend Reverse proxy Upgrade

request Upgrade request HTTP/1.1 101 HTTP/1.1 101 WebSocket connection direct WebSocket connection Client - Backend Backend /socket.io/

-

Smuggling through WebSocket connection 33 Client Frontend Reverse proxy (vulnerable)

Private REST API Public WebSocket API Backend /internal /socket.io/

-

34 Backend Client Frontend Reverse proxy (vulnerable) /internal Upgrade request

/socket.io/ Sec-WebSocket-Version: 1337 Upgrade request Sec-WebSocket-Version: 1337 Version correctness isn't checked! Smuggling through WebSocket connection

-

35 Backend Client Frontend Reverse proxy (vulnerable) /internal Upgrade request

/socket.io/ Sec-WebSocket-Version: 1337 Upgrade request Sec-WebSocket-Version: 1337 HTTP/1.1 426 HTTP/1.1 426 Response correctness isn't checked! Smuggling through WebSocket connection

-

36 Backend Client Frontend Reverse proxy (vulnerable) /internal Upgrade request

/socket.io/ Sec-WebSocket-Version: 1337 Upgrade request Sec-WebSocket-Version: 1337 HTTP/1.1 426 HTTP/1.1 426 TLS connection direct TLS connection Client – Backend not WebSocket!!! Client can access /internal Smuggling through WebSocket connection

-

Challenge – challenge.0ang3el.tk 37 URL <https://challenge.0ang3el.tk/websocket.html> You

need to access flag on localhost:5000 Seems no one solved

-

Challenge – challenge.0ang3el.tk 38 Frontend Not disclosed WebSocket

reverse proxy socket.io.js Proxies only WebSocket API - /socket.io/ path Backend Flask, Flask-SocketIO, Flask-Restful Listens on localhost:5000 only

-

challenge1.py

-

challenge1.py - DEMO

Vulnerable reverse proxies 41 Vulnerable Varnish, Envoy proxy

<= 1.8.0, other non-disclosed Not vulnerable Nginx, HAProxy, Traefik, others

Varnish response 42 WebSocket proxying configuration <https://varnish-cache.org/docs/6.3/users-guide/vcl-example-websockets.html>

Smuggling through WebSocket connection 43 Client Frontend Reverse proxy (Nginx)

or another) Private REST API Public WebSocket API & REST API Backend /internal /api/socket.io/ /api/health

Smuggling through WebSocket connection 44 Client Frontend Reverse proxy (Nginx)

or another) Backend /internal /api/socket.io/ /api/health example.com GET HTTP/1.1 200

Smuggling through WebSocket connection 45 Client Frontend Reverse proxy (Nginx)

or another) Backend /internal /api/socket.io/ /api/health Only Upgrade: websocket header is checked! POST /api/health?u= POST /api/health?u=

Smuggling through WebSocket connection 46 Client Frontend Reverse proxy (Nginx)

or another) Backend /internal /api/socket.io/ /api/health attacker.com GET HTTP/1.1 101 HTTP/1.1 101 HTTP/1.1 101 Only status code is checked for response! POST /api/health?u= POST /api/health?u=

Smuggling through WebSocket connection 47 Client Frontend Reverse proxy (Nginx)

or another) Backend /internal /api/socket.io/ /api/health HTTP/1.1 101 HTTP/1.1 101 TLS connection direct TLS connection Client – Backend not WebSocket!!! Client can access /internal

POST /api/health?u= POST /api/health?u= Client-to-Server masking isn't checked by proxy!!!

-

Challenge2 – challenge2.0ang3el.tk 48 URL <https://challenge2.0ang3el.tk/websocket.html> You

need to access flag on localhost:5000 Seems no one solved

-

Challenge2 – challenge2.0ang3el.tk 49 Frontend Nginx as WebSocket

reverse proxy socket.io.js Proxies only /api/public path (socket.io and healthcheck) Backend Flask, Flask-SocketIO, Flask-Restful Listens on localhost:5000 only

-

Challenge2 – challenge2.0ang3el.tk 50 Nginx config

-

Challenge2 – challenge2.0ang3el.tk 51 REST API - healthcheck

-

Challenge2.py 5

-

Challenge2.py - Demo 5

-

Vulnerable reverse proxies 54 Almost all proxies are affected

But exploitation is limited External SSRF is required that returns status code ...

-

55 Discovering WebSocket APIs

-

Discovering WebSocket API 56 Monitor Upgrade requests Analyze

JavaScript files Try to establish WebSocket connection to each URL ...

-

57 Conclusion

-

Ideas for further research 58 Security of WebSocket subprotocols

More smuggling techniques HTTP/2 and WebSocket ...

-

Thank you! @0ang3l

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